ICCV HONOLULU HAWAII
OCT 19-23, 2025

Foundation Models in Visual Anomaly Detection: Advances, Challenges, and Applications

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FM-driven Image Anomaly Detection

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Outline

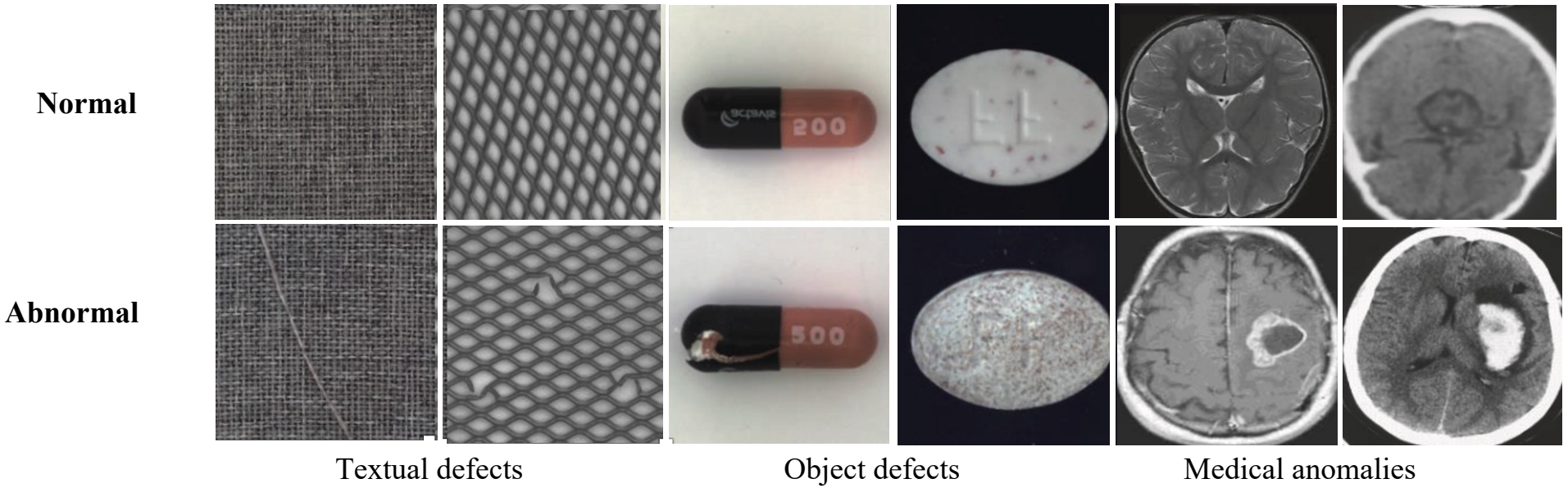
- **Background & Motivation**
- **Simple Training-free & Projection Layer Approaches**
- **Text Prompt Tuning Approaches**
- **Few-shot Residual Learning & Regression Approaches**
- **Reasoning & Multimodal Integration Approaches**

Background & Motivation

Image Anomaly Detection (IAD)

Definition: The goal of IAD is detect or locate unusual or rare patterns that deviate from what is considered normal in an image.

ie., **Industrial defective AD**, **Medical image AD**.



Industrial & Medical AD Datasets

Dataset	Subset	Type	Original Training	Original Test	
			Normal	Normal	Anomaly
MVTec AD	Carpet	Texture	280	28	89
	Grid	Texture	264	21	57
	Leather	Texture	245	32	92
	Tile	Texture	230	33	83
	Wood	Texture	247	19	60
	Bottle	Object	209	20	63
	Capsule	Object	219	23	109
	Pill	Object	267	26	141
	Transistor	Object	213	60	40
	Zipper	Object	240	32	119
	Cable	Object	224	58	92
	Hazelnut	Object	391	40	70
	Metal_nut	Object	220	22	93
	Screw	Object	320	41	119
	Toothbrush	Object	60	12	30
VisA	candle	Object	900	100	100
	capsules	Object	542	60	100
	cashew	Object	450	50	100
	chewinggum	Object	453	50	100
	fryum	Object	450	50	100
	macaroni1	Object	900	100	100
	macaroni2	Object	900	100	100
	pcb1	Object	904	100	100
	pcb2	Object	901	100	100
	pcb3	Object	905	101	100
	pcb4	Object	904	101	100
	pipe_fryum	Object	450	50	100

Data type	Dataset	Modalities	|C|	Normal	Anomalous
Object	SDD	Photography	1	286	54
	BTAD	Photography	3	451	290
	MPDD	Photography	6	176	282
Textual	AITEX	Photography	12	564	183
	DAGM	Photography	10	6996	1054
	DTD-Synthetic	Photography	12	357	947
	ELPV	Electroluminescence	2	377	715
Brain	BrainMRI	Radiology (MRI)	1	98	155
	HeadCT	Radiology (CT)	1	100	100
	Br35H	Radiology (MRI)	1	1500	1500
Fundus	LAG	Fundus Photography	1	786	1711
Colon	CVC-ColonDB	Endoscopy	1	0	380
	CVC-ClinicDB	Endoscopy	1	0	612
	Kvasir	Endoscopy	1	0	1000
	Endo	Endoscopy	1	0	200
Skin	ISIC	Photography	1	0	379
Thyroid	TN3K	Radiology (Utralsound)	1	0	614

Image Anomaly Detection

Definition: The goal of IAD is detect or locat unusual or rare patterns that deviate from what is considered normal in an image.

ie., **AD in semantic level**.

We also consider the anomalies in the semantic level. **Semantic image AD** datasets can be constructed from multi-class classification datasets (*eg*., MNIST, CIFAR-10, *etc*.).

Normal

Abnormal

Semantical anomalies

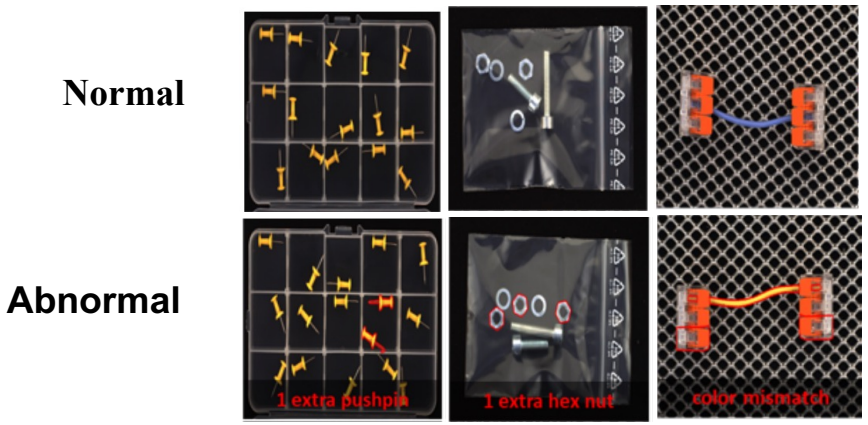
Dataset	Subset	Type	Test set	
			Normal	Anomaly
MNIST	0	Semantical	980	9020
	1	Semantical	1135	8865
	2	Semantical	1,032	8,968
	3	Semantical	1,010	8,990
	4	Semantical	982	9019
	5	Semantical	892	9108
	6	Semantical	958	9042
	7	Semantical	1028	8972
	8	Semantical	974	9026
	9	Semantical	1009	8991
	even_number	Semantical	4926	5074
CIFAR-10	airplane	Semantical	1000	9000
	automobile	Semantical	1000	9000
	bird	Semantical	1000	9000
	cat	Semantical	1000	9000
	deer	Semantical	1000	9000
	dog	Semantical	1000	9000
	frog	Semantical	1000	9000
	horse	Semantical	1000	9000
	ship	Semantical	1000	9000
	truck	Semantical	1000	9000
	animal	Semantical	6000	4000

Image Anomaly Detection

Definition: The goal of IAD is detect or locat unusual or rare patterns that deviate from what is considered normal in an image.

ie., **AD in semantic level**.

Logical image AD aims to identify anomalies that violate high-level semantic rules in images, such as incorrect quantities, arrangements, or component combinations.



Dataset	Subset	Type	Test set	
			Normal	Anomaly
MVTec LOCO AD	screw_bag	Logical	102	173
	pushpins	Logical	122	219
	splicing_connectors	Logical	138	172
	juice_bottle	Logical	119	193
	breakfast_box	Logical	94	236

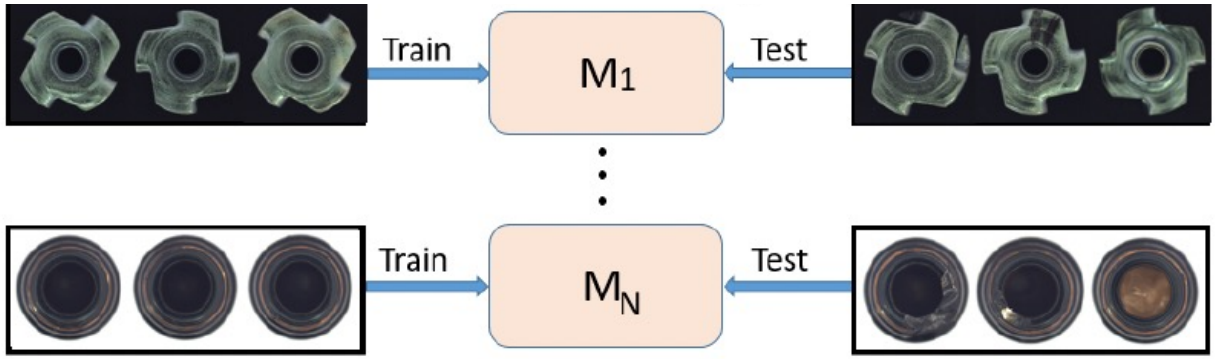
Logical anomalies

Why Foundation Models (FMs) for Image Anomaly Detection (IAD)?

Limitations of traditional IAD approaches

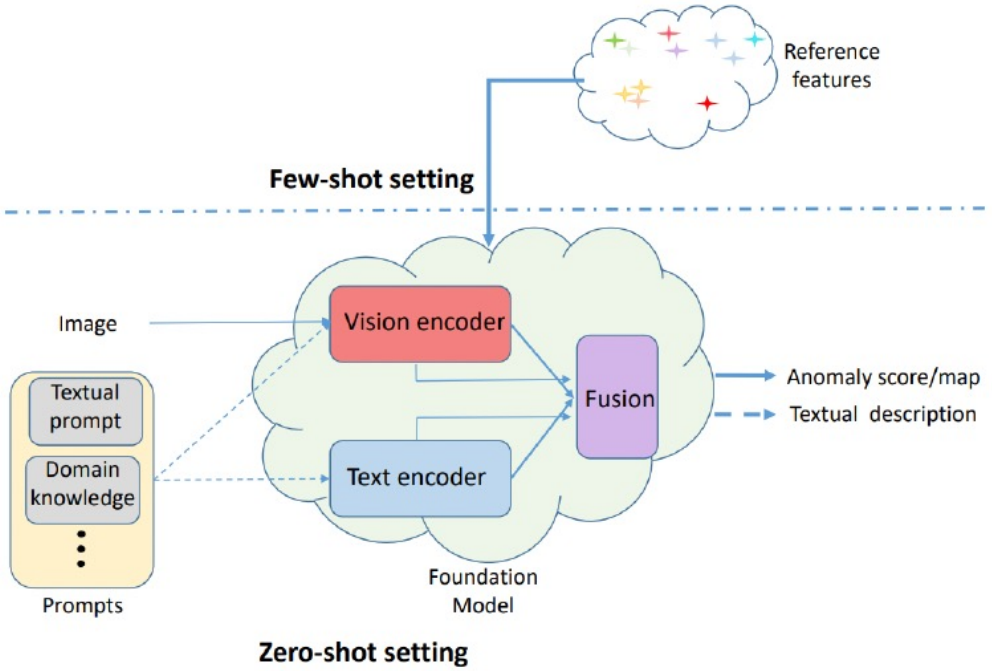
- Conventional anomaly detection methods usually rely on domain-specific training and require large collections of normal samples.
- Their performance often deteriorates when deployed in unseen domains, due to poor generalization beyond the training distribution.

One-for-one setting



Why Foundation Models (FMs) for Image Anomaly Detection (IAD)?

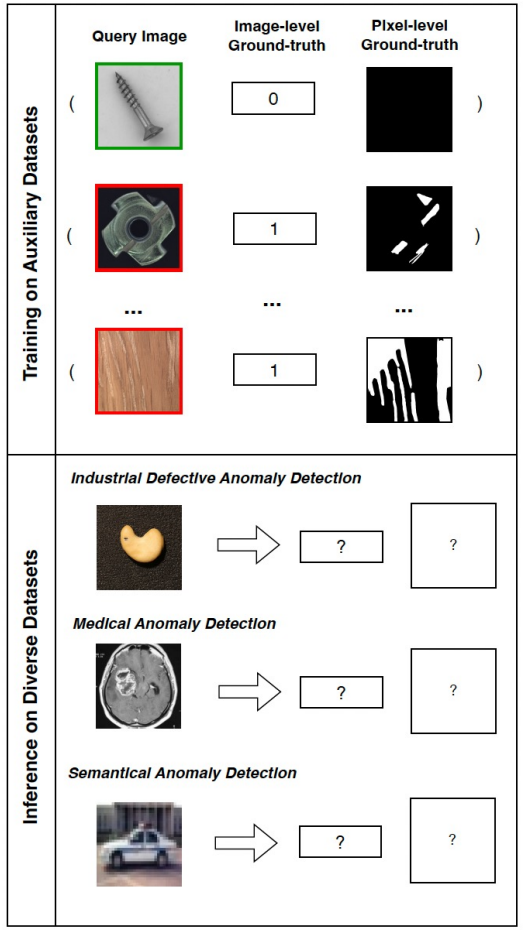
Rich priors in FMs:
Large Foundation Models are pretrained on massive web-scale data, capturing broad visual and/or semantic knowledge. These priors enable strong cross-domain generalization and reduce the need for task-specific retraining.



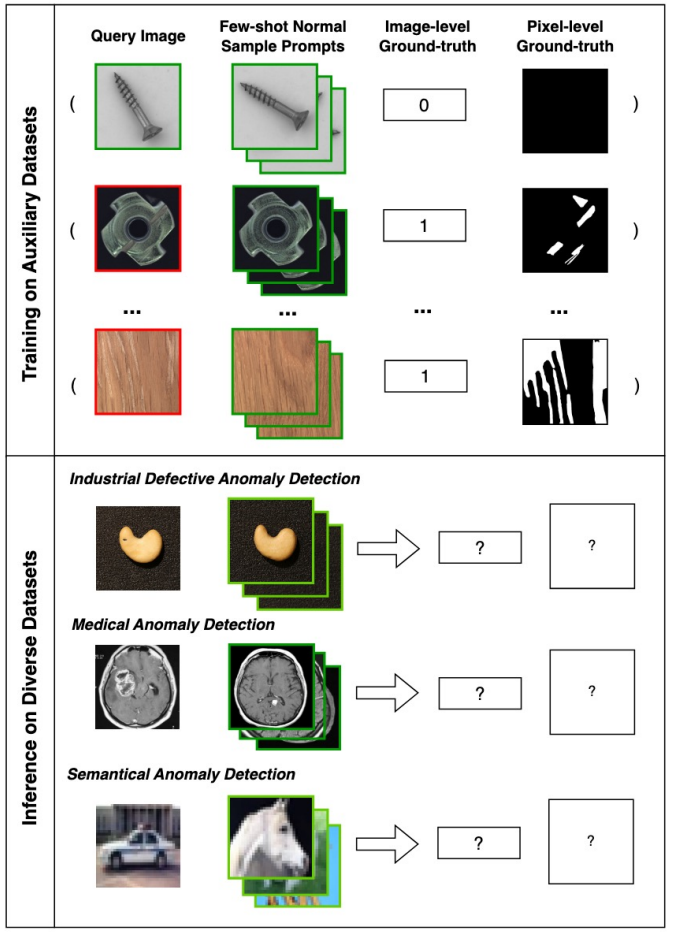
- **Visual Models:** SAM, DINO/DINOv2, Diffusion Models, etc.
- **Visual-Language Models:** **CLIP**, ImageBind-Huge, etc.
- **Language Models:** GPT4/GPT4v, LLaMA, etc.

Foundation Model-driven Image Anomaly Detection

Zero-shot Anomaly Detection



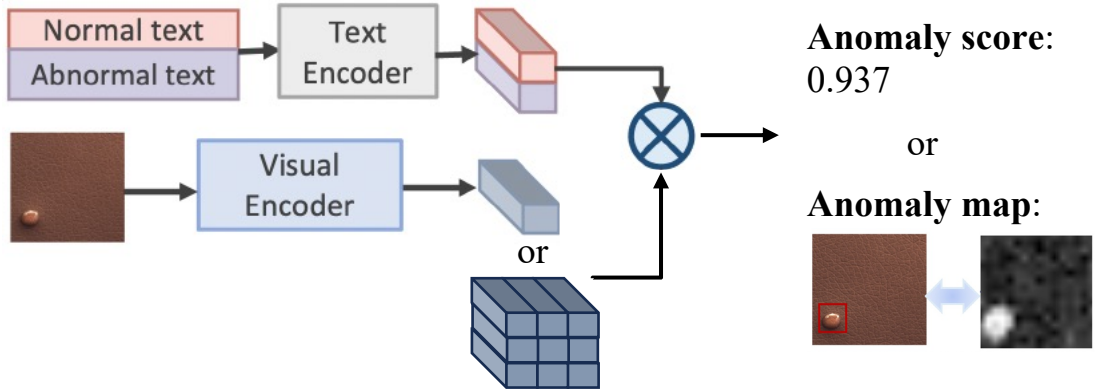
Few-shot Anomaly Detection



* Typically, MVTecAD is the default auxiliary dataset.

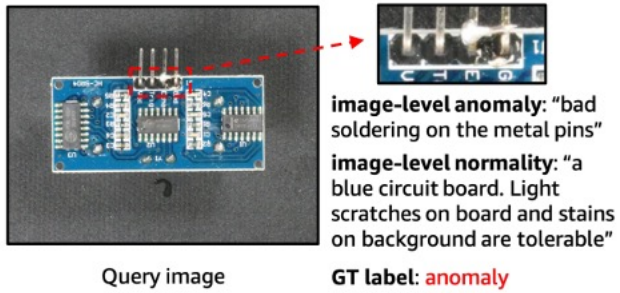
General Learning Paradigm of VLM-based IAD

Zero-shot AD Scoring



In **zero-shot learning**, the anomaly scoring is often based on the similarity between text features and visual features.

Handcrafted Text Prompt



Learnable Text Prompt

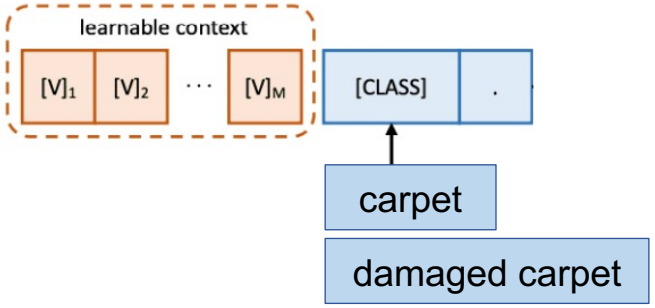
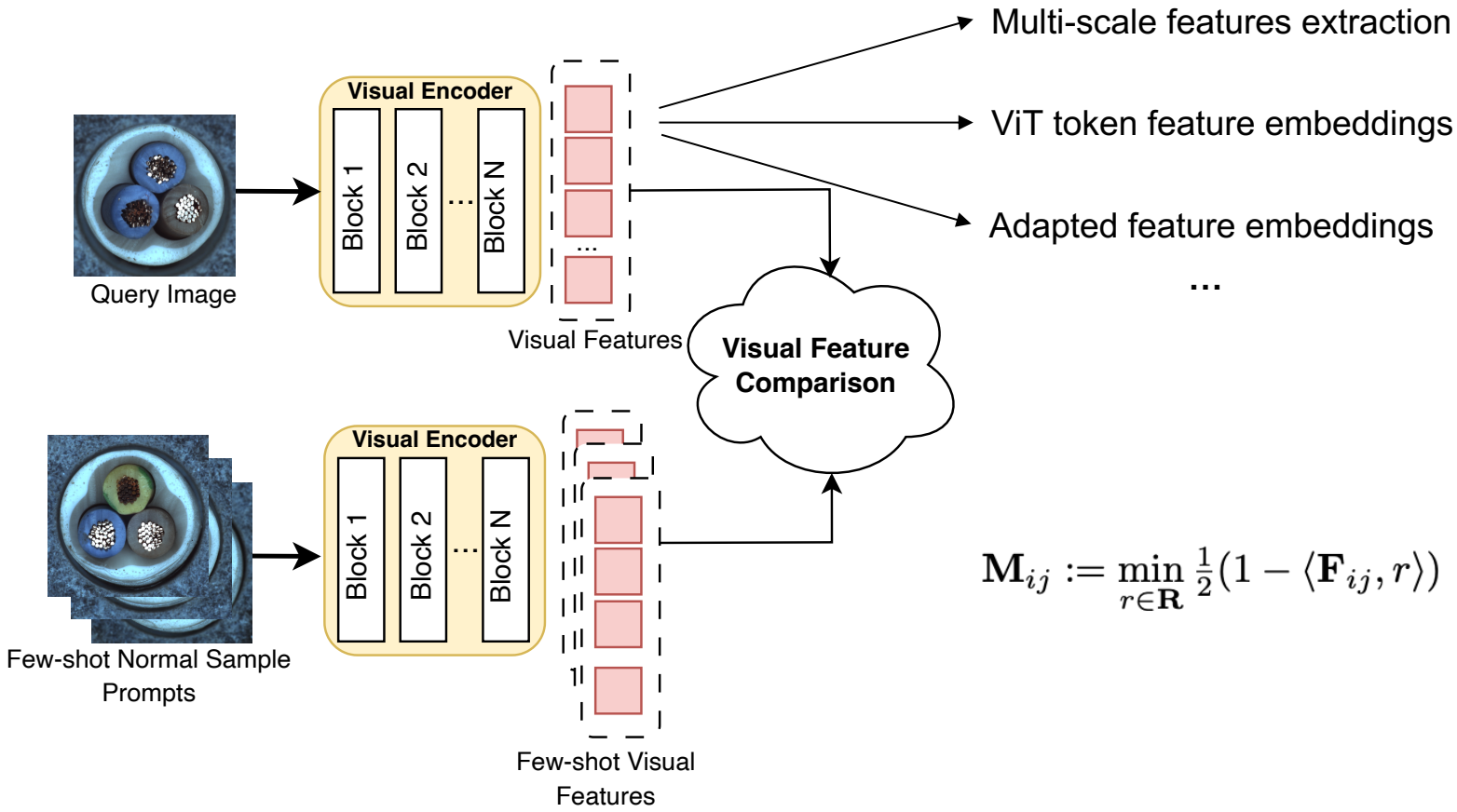


Image-level score: $$s_a(x) = \frac{\exp(\mathbf{F}_a^\intercal f_v(x))}{\exp(\mathbf{F}_n^\intercal f_v(x)) + \exp(\mathbf{F}_a^\intercal f_v(x))}$$

Pixel-level score: $$\mathbf{S}_x^a(i,j) = \frac{\exp(\mathbf{F}_v(i,j)\mathbf{F}_a^\intercal)}{\exp(\mathbf{F}_v(i,j)\mathbf{F}_n^\intercal) + \exp(\mathbf{F}_v(i,j)\mathbf{F}_a^\intercal)}$$

General Learning Paradigm of VLM-based IAD

In **few-shot learning**, we usually build a feature memory bank that stores the visual features of these few-shot normal samples**.**
The model then computes the similarity between the query feature and features* from the memory bank to determine how "normal" or "abnormal" the image/patch is.



* We usually select the most similar feature as the query feature from the feature memory bank.

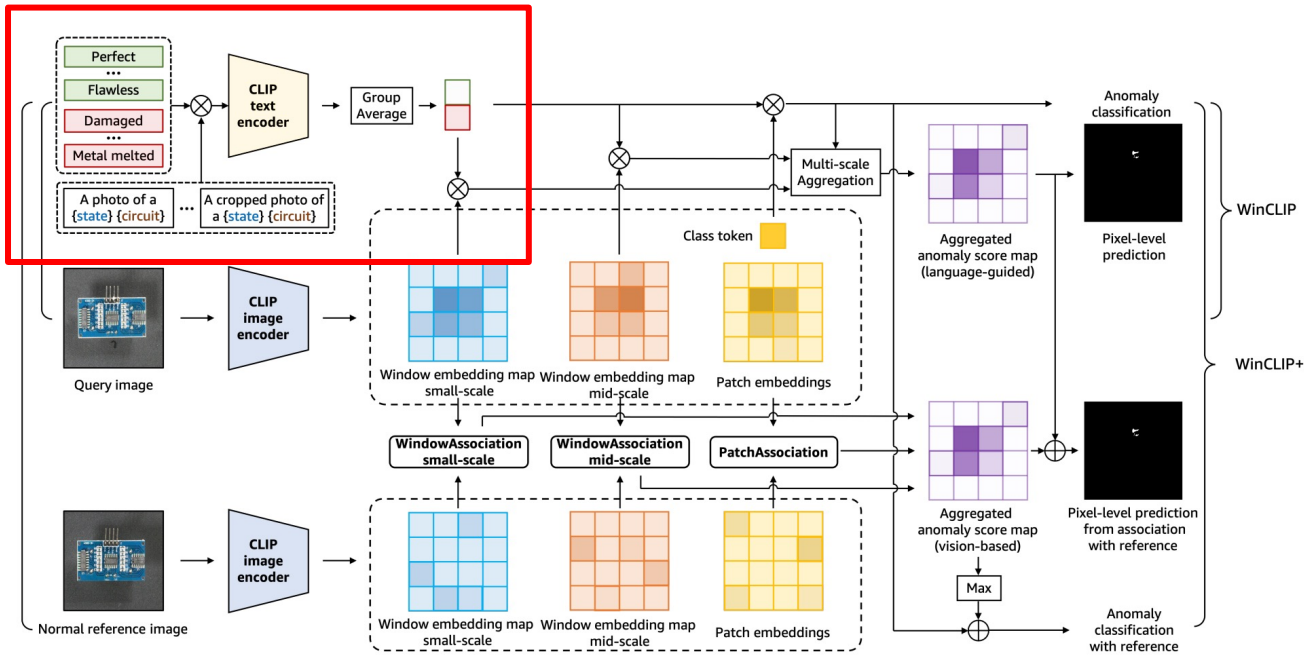
Taxonomy of FM-driven IAD Approaches

Categories	Method	Venue	Setup	Foundation Models
Simple Training-free & Projection Layer	WinCLIP	CVPR'23	Zero- /Few-shot	CLIP
	AnoVL	Arxiv	Zero-shot	CLIP
	AprilGAN	CVPR'23 (VAND)	Zero- /Few-shot	CLIP
Text Prompt Tuning	AnomalyCLIP	ICLR'24	Zero-shot	CLIP
	AdaCLIP	ECCV'24	Zero-shot	CLIP
	FiLo	MM'24	Zero-shot	CLIP, Grounding DINO, LLM
	AA-CLIP	CVPR'25	Zero-shot	CLIP
	Bayes-PFL	CVPR'25	Zero-shot	CLIP
	Mao et. al.	CVPR'25	Zero- /Few-shot	CLIP
	FAPrompt	ICCV'25	Zero-shot	CLIP
Few-shot Residual Learning & Regression	InCTRL	CVPR'24	Few-shot	CLIP
	ResAD	Neurips'24	Few-shot	CLIP, ImageBind-Huge
	One-to-Normal	Neurips'24	Few-shot	CLIP, Stable Diffusion V1.5
Reasoning & Multimodal Intergration	AnomalyGPT	AAAI'24	Few-shot	CLIP, ImageBind-Huge, Vicuna-7B
	Triad	CVPR'25	Zero- /Few-shot	LLaVA-1.6, LLaVA-OneVision
	Anomaly-OV	CVPR'25	Zero-shot	CLIP, LLaVAOneVision
	LogSAD	CVPR'25	Few-shot	SAM, CLIP, DINOv2, GPT-4V

Simple Training-free & Projection Layer

Training-free Baseline: WinCLIP

WinCLIP computes the average of text embeddings for all normal and abnormal prompts separately, representing the **normal** and **abnormal classes** in the feature space.



$$\mathbf{F}_n = \frac{1}{|\mathcal{P}_t^n|} \sum_{p_i \in \mathcal{P}_t^n} f_t(p_i) \qquad \mathbf{F}_a = \frac{1}{|\mathcal{P}_t^a|} \sum_{p_j \in \mathcal{P}_t^a} f_t(p_j)$$

Hand-crafted Text Prompt

(a) *State*-level (normal)

- `c := "[o]"`
- `c := "flawless [o]"`
- `c := "perfect [o]"`
- `c := "unblemished [o]"`
- `c := "[o] without flaw"`
- `c := "[o] without defect"`
- `c := "[o] without damage"`

(b) *State*-level (anomaly)

- `c := "damaged [o]"`
- `c := "[o] with flaw"`
- `c := "[o] with defect"`
- `c := "[o] with damage"`

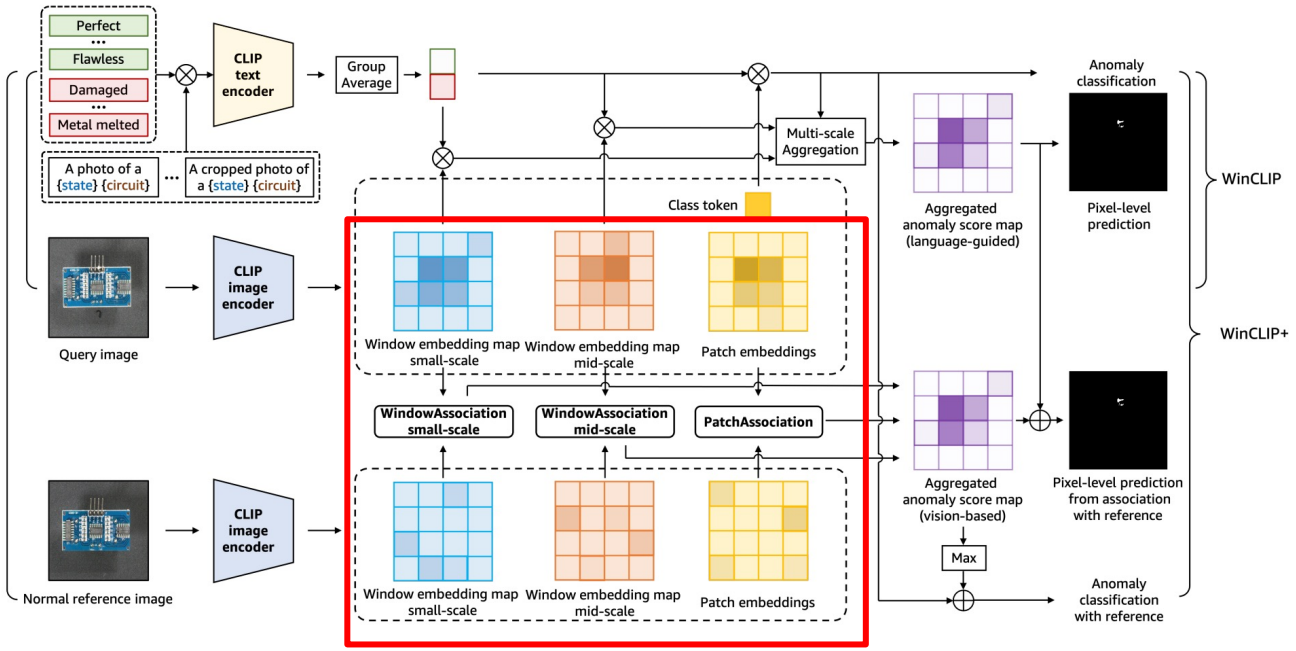
(c) *Template*-level

- `"a cropped photo of the [c]."`
- `"a cropped photo of a [c]."`
- `"a close-up photo of a [c]."`
- `"a close-up photo of the [c]."`
- `"a bright photo of a [c]."`
- `"a bright photo of the [c]."`
- `"a dark photo of the [c]."`
- `"a dark photo of a [c]."`
- `"a jpeg corrupted photo of a [c]."`
- `"a jpeg corrupted photo of the [c]."`

Normal Examples	*'a photo of a flawless [c] for visual inspection.'* *'a cropped photo of a perfect [c].'* *'a blurry photo of the [c] without defect.'* *'a dark photo of the unblemished [c].'* *'a jpeg corrupted photo of a [c] without flaw.'*
Abnormal Examples	*'a photo of a [c] with flaw for visual inspection.'* *'a cropped photo of a [c] with damage.'* *'a blurry photo of the [c] with defect.'* *'a dark photo of the [c] with flaw.'* *'a jpeg corrupted photo of a [c] with defect.'*

Jeong, Jongheon, et al. "Winclip: Zero-/few-shot anomaly classification and segmentation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.

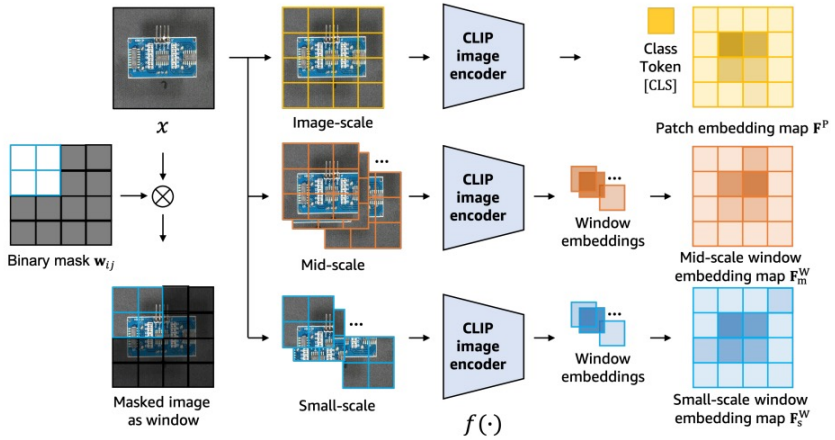
Training-free Baseline: WinCLIP



Each window feature is assigned an **anomaly score** based on cosine similarity. That score is **uniformly distributed** to all pixels within the corresponding window region.

$$\mathbf{M}_{ij} := \min_{r \in \mathbf{R}} \tfrac{1}{2}(1 - \langle \mathbf{F}_{ij}, r \rangle) \qquad \mathbf{M}^{\mathrm{W}} := \tfrac{1}{3}(\mathbf{M}^{\mathrm{P}} + \mathbf{M}^{\mathrm{W}}_{\mathrm{s}} + \mathbf{M}^{\mathrm{W}}_{\mathrm{m}})$$

Multi-scale features Extraction



1. It capture image features at different spatial scales by applying **sliding windows of multiple sizes** to the image.
2. Each window activates a local region of the image, and the CLIP image encoder extracts **window-based features** directly.

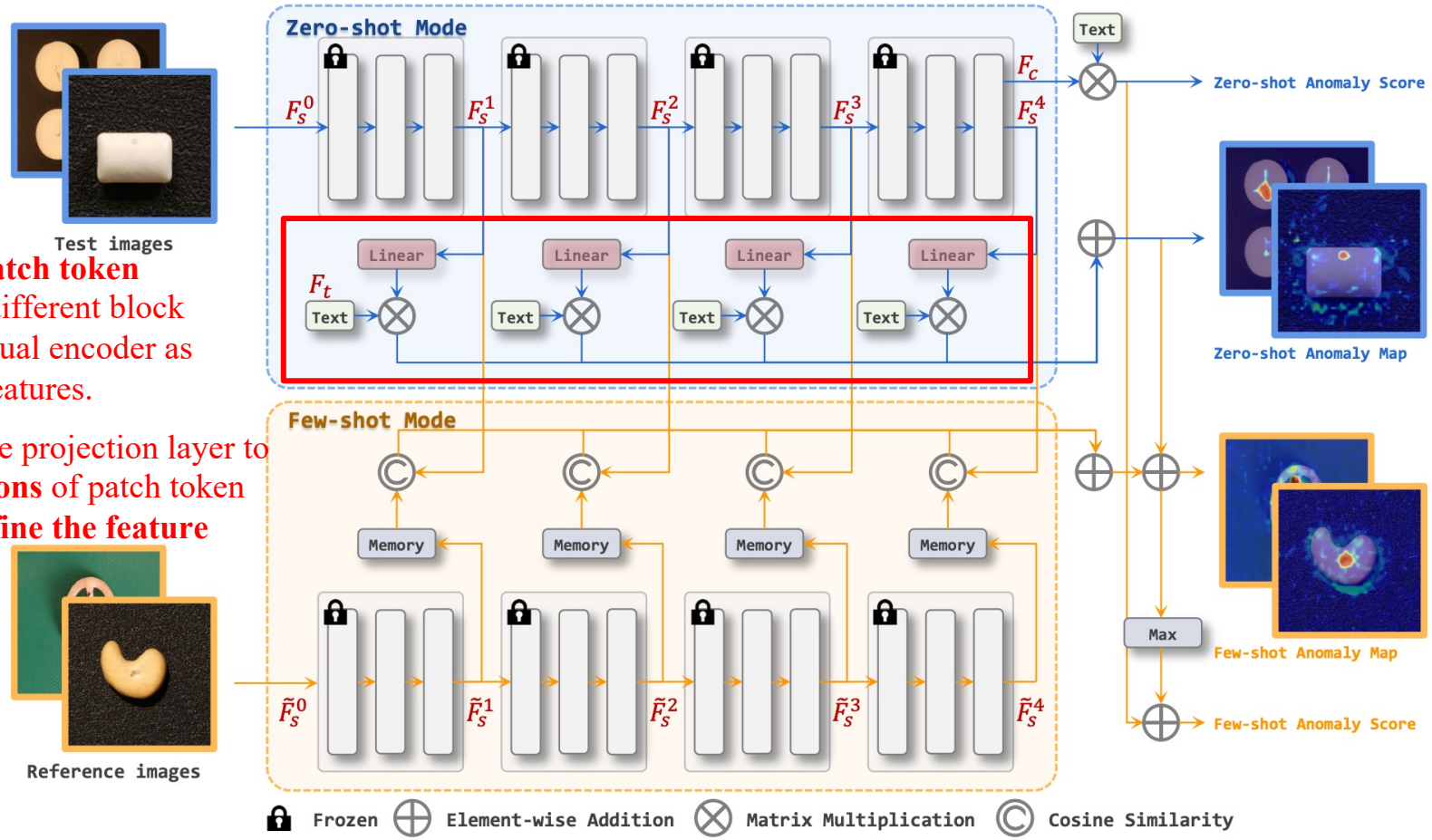
$$\mathbf{F}^{\mathrm{W}}_{ij} := f(\mathbf{x} \odot \mathbf{w}_{ij})$$

Jeong, Jongheon, et al. "Winclip: Zero-/few-shot anomaly classification and segmentation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.

Simple Projection Layer: AprilGAN

1. AprilGAN use **patch token embeddings** from different block layers in CLIP's visual encoder as patch-level visual features.

2. It injects learnable projection layer to **match the dimensions** of patch token embeddings and **refine the feature representations**.

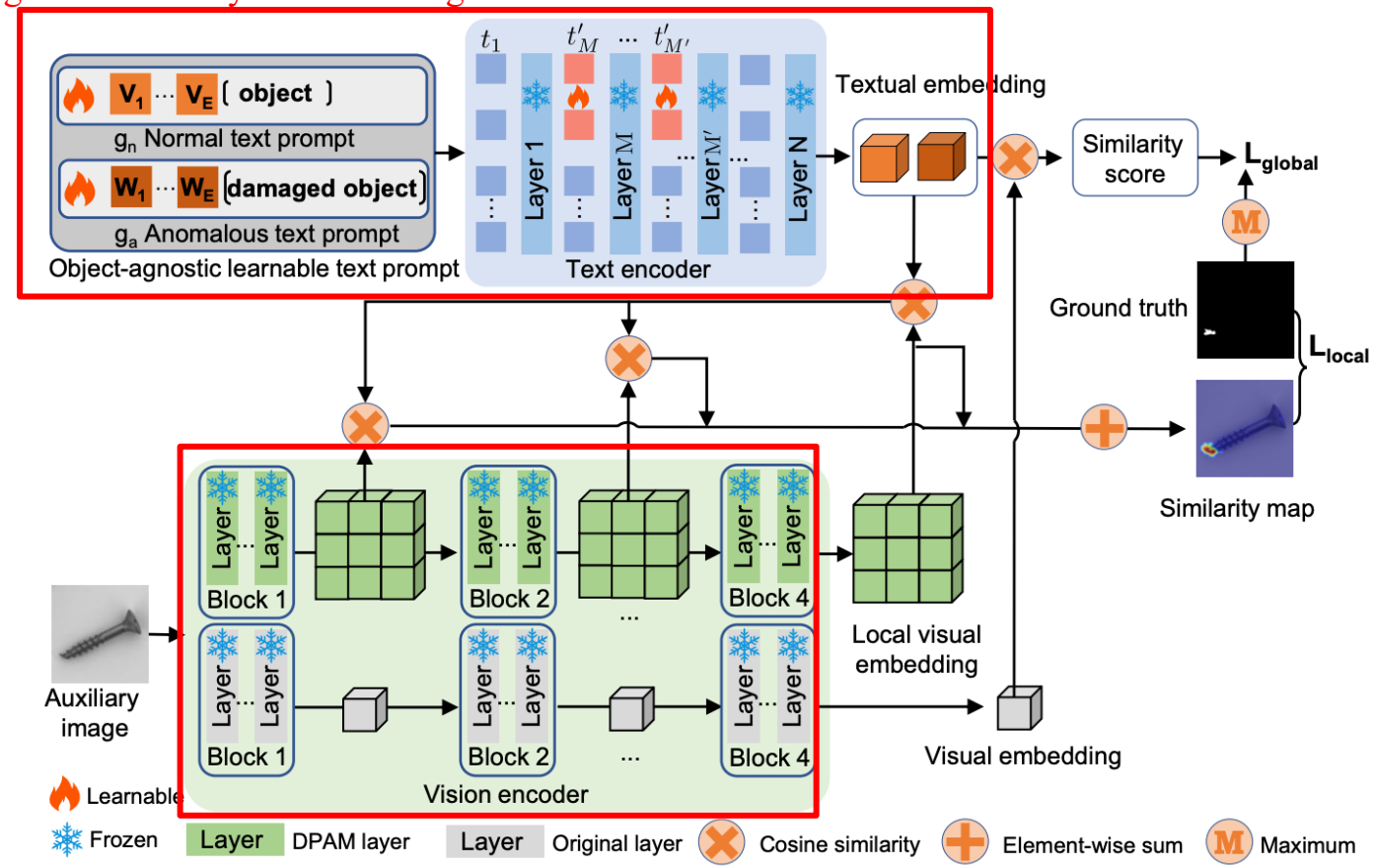


Chen X, Han Y, Zhang J. April-gan: A zero-/few-shot anomaly classification and segmentation method for cvpr 2023 vand workshop challenge tracks 1&2: 1st place on zero-shot ad and 4th place on few-shot ad.

Text Prompt Tuning

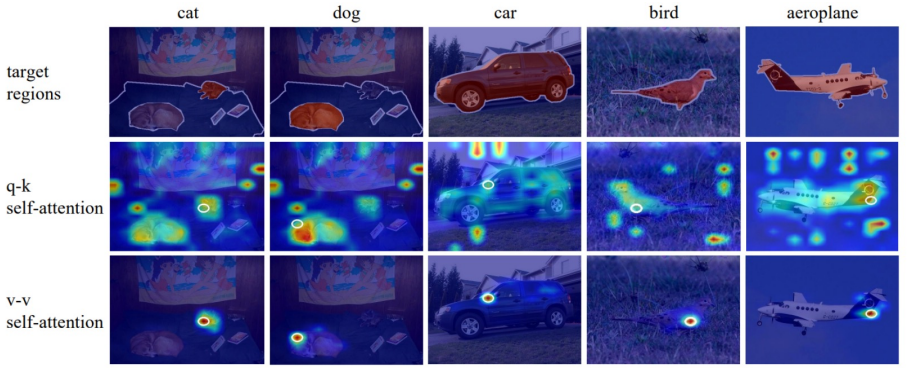
AnomalyCLIP

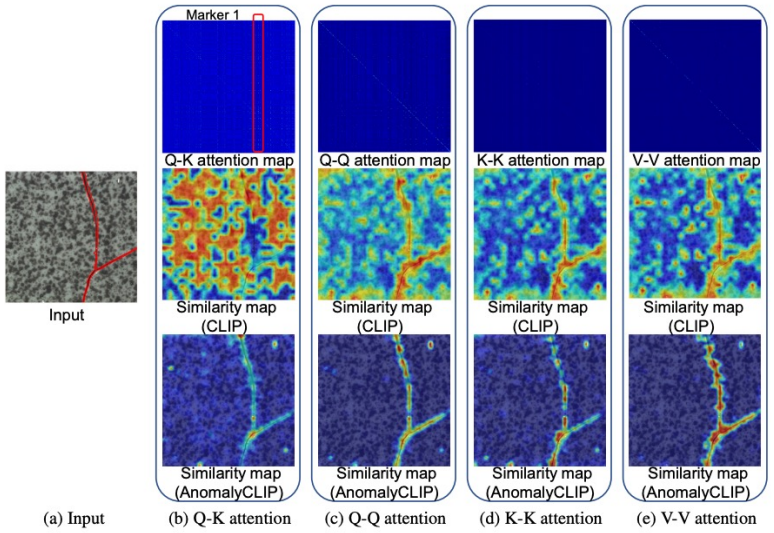
① **Learnable Text Prompts**: a set of learnable tokens + fixed class name token.
This learnable design generalizes better than hand-crafted templates and enables domain-agnostic anomaly understanding



② Diagonally Prominent Attention Map (DPAM) used to refine the local visual space, which can help **reduce the disturbance from other tokens.**

CLIP Surgery





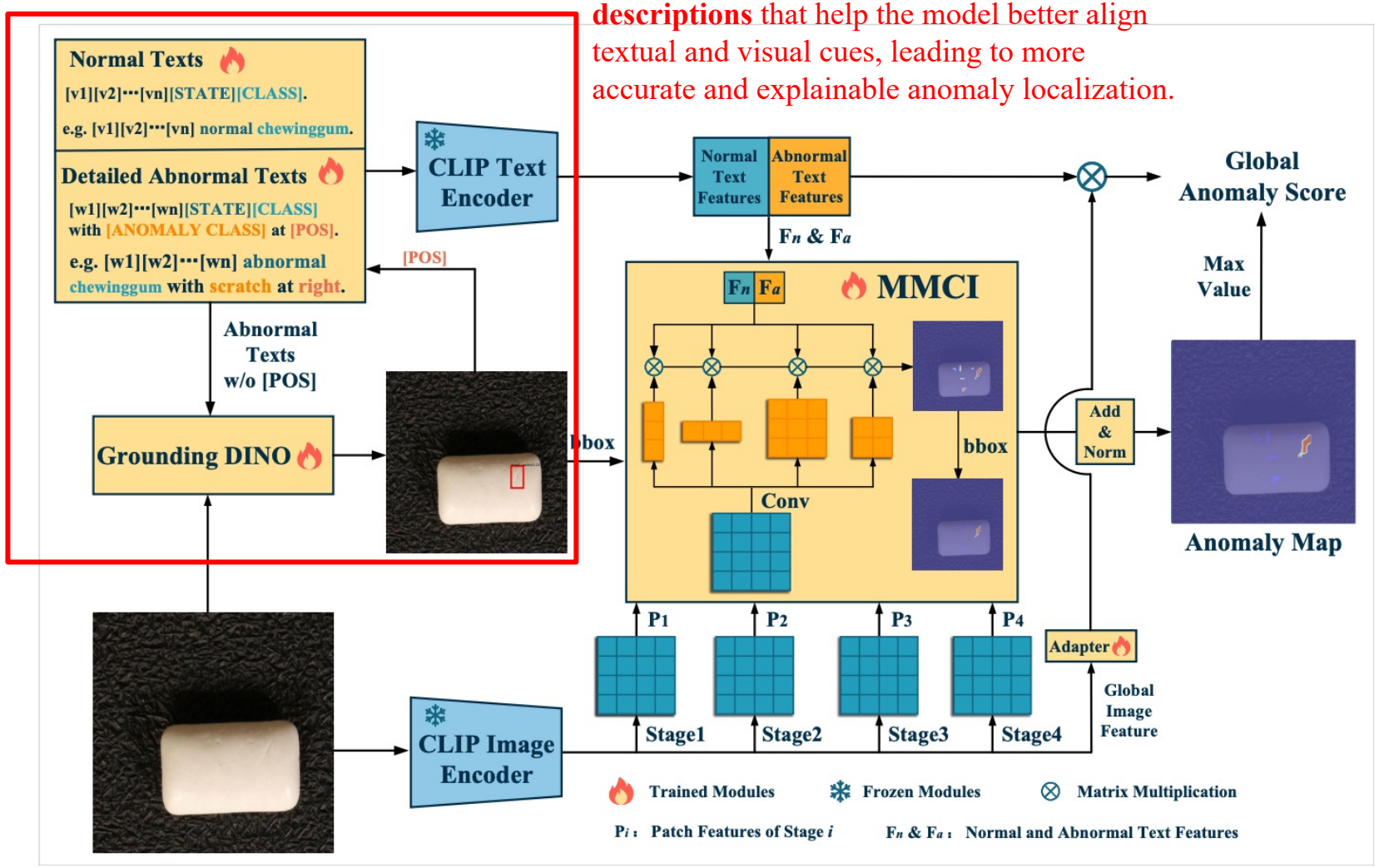
Zhou, Qihang, et al. "Anomalyclip: Object-agnostic prompt learning for zero-shot anomaly detection." The Twelfth International Conference on Learning Representations (2024).

FiLO

[POS]: location of the anomaly region

[ANOMALY CLASS]: potential anomaly types generated by LLM

By combining these two types of contextual information, FiLo creates **fine-grained textual descriptions** that help the model better align textual and visual cues, leading to more accurate and explainable anomaly localization.



Gu, Zhaopeng, et al. "Filo: Zero-shot anomaly detection by fine-grained description and high-quality localization." Proceedings of the 32nd ACM International Conference on Multimedia. 2024

Our approach: FAPrompt

Existing ZSAD methods often design/learn prompts taking inspiration from image classification task.
***e.g., Normal**: ' perfect ' or ' flawless ' carpet*
***Abnormal**: 'damaged' or 'defective' carpet*

They overlook the **intrinsic relationships** between normality and abnormality, where abnormal samples are typically subtle deviations from normal patterns rather than entirely distinct semantic categories.

As a result, the previous methods are prone to learn excessively **simple and homogeneous** abnormal patterns

Motivation of FAPrompt: keep the common features between normality and abnormality, make sure all abnormality prompts are semantically adherent to the normality captured in the normal prompt while introducing learnable deviations to capture fine-grained abnormalities.

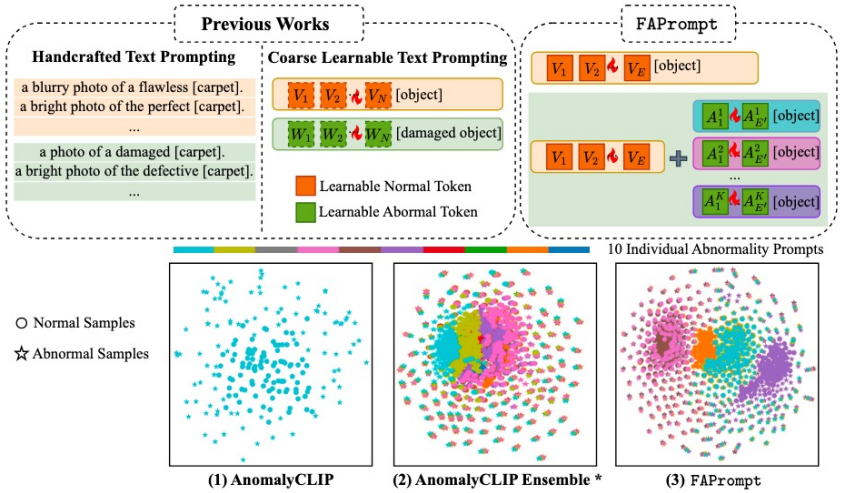
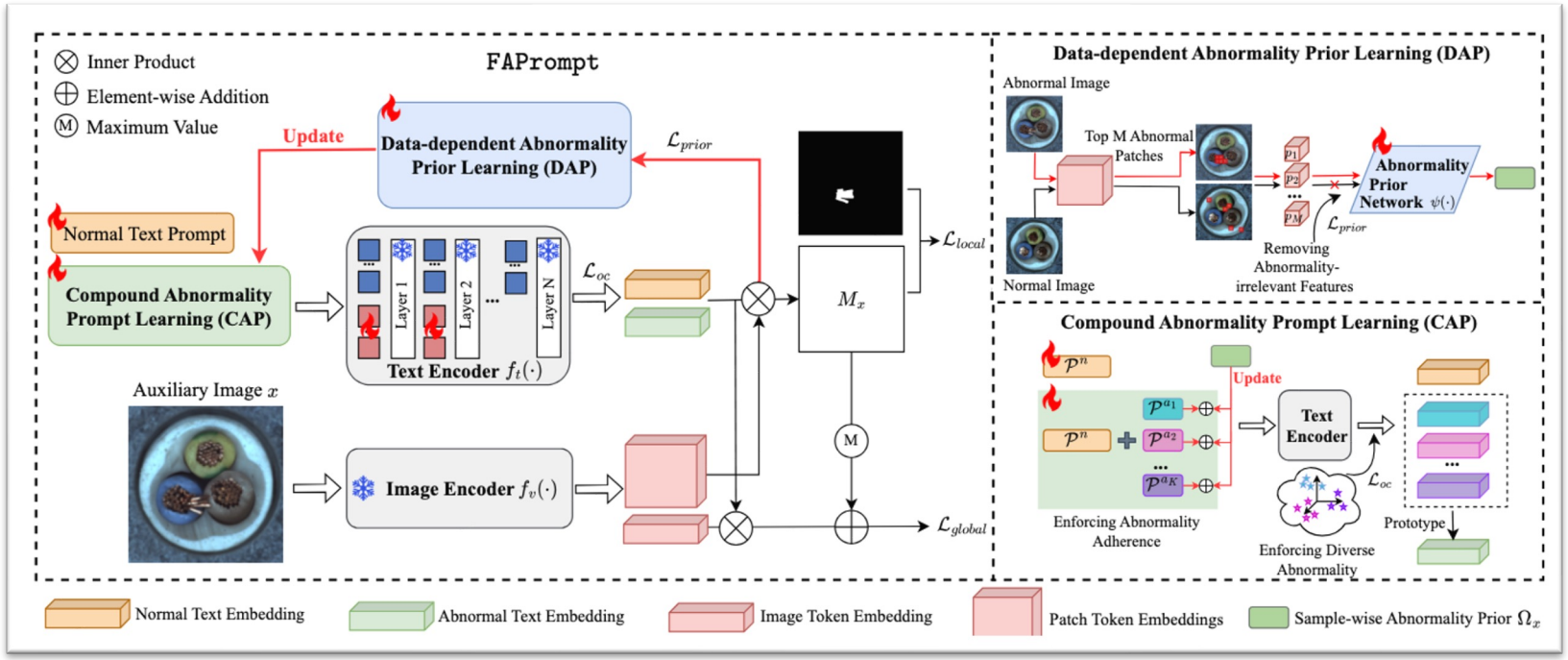


Figure 1. **Top:** FAPrompt vs. previous methods. **Bottom**: 2-D t-SNE visualization of the prompt-wise anomaly score maps of three methods on samples from a BTAD dataset [35], with the maps generated based on the similarities between text prompt embeddings and image embeddings, as defined in Eq. 3.

Our approach: FAPrompt



Adaptively select the most abnormal feature from each query/test image as sample-wise abnormality prior to enhance the cross-domain generalizability of abnormality prompts in CAP.

Abnormality prior learning loss:

$$\mathcal{L}_{prior} = \sum_{y_x=0} \sum_{\omega \in \Omega_x} \omega_x^2,$$

Learn fine-grained abnormalites that semantically align with normality and keep diverse enough to capture different abnormal patterns.

Orthogonal constraint loss:

$$\mathcal{L}_{oc} = \sum_{i,j \in K; i \neq j} abs\left(\frac{f_t(\mathcal{P}^{a_i}) \cdot f_t(\mathcal{P}^{a_j})}{||f_t(\mathcal{P}^{a_i})|| \times ||f_t(\mathcal{P}^{a_j})||}\right),$$

CAP ⟶

$$\mathcal{P}^n = [V_1][V_2]...[V_E][object],$$
$$\mathcal{P}^a = \{\mathcal{P}^{a_1}, \mathcal{P}^{a_2}, ...\mathcal{P}^{a_K}\},$$
$$with\ \mathcal{P}^{a_i} = [V_1][V_2]...[V_E][A_1^i][A_2^i]...[A_{E'}^i][object],$$

DAP ⟶

$$\hat{\mathcal{P}}^a = \{\hat{\mathcal{P}}^{a_1}, \hat{\mathcal{P}}^{a_2}, ...\hat{\mathcal{P}}^{a_K}\}$$
$$with\ \hat{\mathcal{P}}^{a_i} = [V_1][V_2]...[V_E][A_1^i \oplus \Omega_x][A_2^i \oplus \Omega_x]...[A_{E'}^i \oplus \Omega_x][object],$$

Zhu, Jiawen, et al. "Fine-grained Abnormality Prompt Learning for Zero-shot Anomaly Detection." ICCV (2025).

Our approach: FAPrompt

Data Type	Dataset	Handcrafted Text Prompts					Learnable Text Prompts					
		CLIP	CLIP-AC	WinCLIP	APRIL-GAN	AnoVL	CoOp	CoCoOp	BLIP	AnomalyCLIP	FiLO	FAPrompt
Industrial	MVTecAD	(74.1, 87.6)	(71.5, 86.4)	(91.8, **96.5**)	(86.2, 93.5)	(**92.5**, **96.7**)	(88.8, 94.8)	(71.8, 84.9)	(72.3, 86.7)	(91.5, 96.2)	(91.2, 96.0)	(**91.9**, 95.7)
	VisA	(66.4, 71.4)	(65.0, 70.2)	(78.8, 81.4)	(78.0, 81.4)	(79.2, 81.7)	(62.8, 68.1)	(78.1, 82.3)	(57.8, 64.6)	(82.1, 84.6)	(**83.9**, **87.3**)	(**84.6**, **86.8**)
	BTAD	(34.5, 52.5)	(51.0, 62.1)	(68.2, 70.9)	(73.6, 68.6)	(80.3, 73.1)	(66.8, 77.4)	(48.4, 53.9)	(85.3, 91.2)	(**88.3**, 87.3)	(85.4, **91.5**)	(**92.2**, **92.5**)
	AITEX	(71.0, 45.7)	(71.5, 46.7)	(73.0, 54.7)	(57.6, 41.3)	(72.5, 55.4)	(66.2, 39.0)	(48.6, 37.8)	(71.3, 52.2)	(62.2, 40.4)	(**78.2**, **59.9**)	(**74.1**, **55.5**)
	DAGM	(79.6, 59.0)	(82.5, 63.7)	(91.8, 79.5)	(94.4, 83.8)	(89.7, 76.3)	(87.5, 74.6)	(96.3, 85.5)	(84.3, 66.6)	(**97.5**, **92.3**)	(96.6, 90.4)	(**98.8**, **95.3**)
	DTD-Synthetic	(71.6, 85.7)	(66.8, 83.2)	(93.2, 92.6)	(86.4, 95.0)	(**94.9**, 97.3)	(83.1, 91.9)	(84.1, 92.9)	(91.5, 96.3)	(93.5, 97.0)	(94.7, **98.0**)	(**96.2**, **98.1**)
	ELPV	(59.2, 71.7)	(69.4, 80.2)	(74.0, 86.0)	(65.5, 79.3)	(70.6, 83.0)	(73.0, 86.5)	(78.4, 89.2)	(75.2, 87.3)	(81.5, **91.3**)	(**82.2**, 91.2)	(**83.7**, **92.1**)
	SDD	(95.5, 87.9)	(94.7, 77.9)	(94.0, 87.2)	(97.5, 93.4)	(95.3, 91.3)	(96.8, 90.0)	(89.9, 50.4)	(85.8, 56.9)	(**98.1**, 93.4)	(97.9, **94.8**)	(**98.4**, **95.6**)
	MPDD	(54.3, 65.4)	(56.2, 66.0)	(63.6, 69.9)	(73.0, 80.2)	(68.9, 71.9)	(55.1, 64.2)	(61.0, 69.1)	(59.5, 67.7)	(**77.0**, **82.0**)	(74.4, 76.9)	(**80.1**, **83.9**)
Medical	BrainMRI	(73.9, 81.7)	(80.6, 86.4)	(86.6, 91.5)	(89.3, 90.9)	(88.7, 91.3)	(61.3, 44.9)	(78.2, 86.7)	(75.7, 83.0)	(90.3, 92.2)	(**94.5**, **94.9**)	(**95.8**, **96.2**)
	HeadCT	(56.5, 58.4)	(60.0, 60.7)	(81.8, 80.2)	(89.1, 89.4)	(81.6, 84.2)	(78.4, 78.8)	(80.3, 73.4)	(68.1, 62.8)	(93.4, 91.6)	(**93.9**, **92.1**)	(**94.0**, **92.4**)
	LAG	(58.7, 76.5)	(58.2, 76.9)	(59.2, 74.8)	(73.6, 84.8)	(65.1, 78.0)	(69.6, 82.9)	(72.6, 84.7)	(48.3, 66.8)	(**74.3**, **84.9**)	(71.3, 83.8)	(**76.6**, **86.1**)
	Br35H	(78.4, 78.8)	(82.7, 81.3)	(80.5, 82.2)	(93.1, 92.9)	(88.4, 88.9)	(86.0, 87.5)	(85.7, 89.1)	(76.4, 74.4)	(94.6, 94.7)	(**97.7**, **96.8**)	(**97.6**, **97.1**)

Table 1. Image-level ZSAD results (AUROC, AP) on 13 AD datasets. The best and second-best results are respectively highlighted in **red** and **blue**. The results for MVTecAD, VisA, DAGM, DTD-Synthetic, BTAD, and MPDD are averaged performance across their multiple data subsets (see Appendix D.1 for breakdown results).

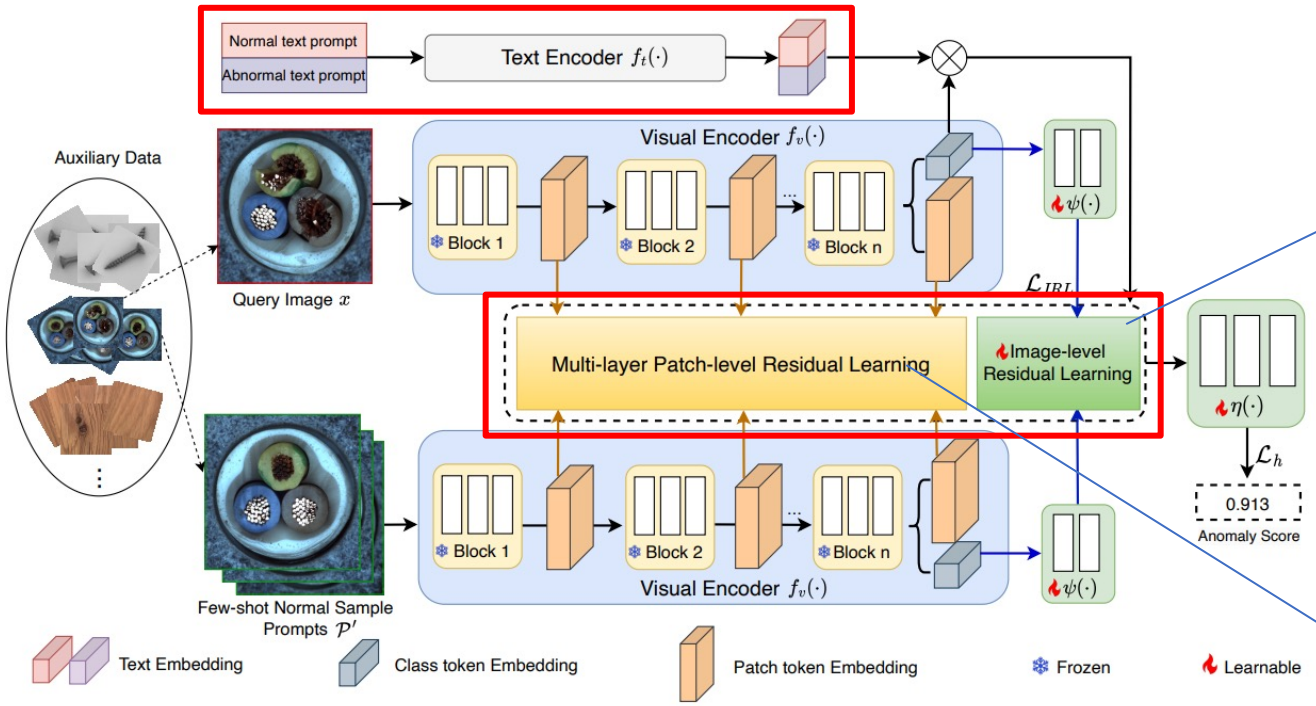
Data Type	Dataset	Handcrafted Text Prompts					Learnable Text Prompts					
		CLIP	CLIP-AC	WinCLIP	APRIL-GAN	AnoVL	CoOp	CoCoOp	BLIP	AnomalyCLIP	FiLO	FAPrompt
Industrial	MVTecAD	(38.4, 11.3)	(38.2, 11.6)	(85.1, 64.6)	(87.6, 44.0)	(89.8, 76.2)	(33.3, 6.6)	(86.7, 79.6)	(85.1, 68.2)	(**91.1**, **81.4**)	(**92.3**, 61.7)	(90.6, **83.3**)
	VisA	(46.6, 14.8)	(47.8, 17.2)	(79.6, 56.8)	(94.2, 86.8)	(89.9, 71.2)	(24.1, 3.8)	(93.6, 86.7)	(85.7, 62.4)	(**95.5**, **87.0**)	(**95.9**, 85.4)	(**95.9**, **87.7**)
	BTAD	(30.6, 4.4)	(32.8, 8.3)	(72.7, 27.3)	(60.8, 25.0)	(93.2, 62.8)	(28.6, 3.8)	(86.1, 72.0)	(89.3, 60.7)	(94.2, **74.8**)	(**94.5**, 56.3)	(**95.6**, **75.1**)
	AITEX	(53.2, 15.3)	(47.3, 11.8)	(62.5, 41.5)	(78.2, **68.8**)	(59.2, 49.1)	(67.7, 54.9)	(52.1, 56,9)	(59.0, 57.7)	(**83.0**, **66.5**)	(70.7, 25.3)	(**82.0**, 66.2)
	DAGM	(28.2, 2.9)	(32.7, 4.8)	(87.6, 65.7)	(82.4, 66.2)	(92.0, 78.8)	(17.5, 2.1)	(82.8, 75.1)	(90.0, 78.0)	(95.6, **91.0**)	(**96.8**, 90.6)	(**98.2**, **95.0**)
	DTD-Synthetic	(33.9, 12.5)	(23.7, 5.5)	(83.9, 57.8)	(95.3, 86.9)	(97.5, 90.4)	(55.8, 36.0)	(93.7, 83.7)	(96.3, 88.2)	(97.9, **92.3**)	(**98.1**, 88.6)	(**98.3**, **93.3**)
	SDD	(28.4, 5.1)	(33.5, 7.6)	(95.9, 78.4)	(93.0, 84.6)	(97.9, 82.6)	(91.8, 81.7)	(93.7, 85.0)	(95.6, 76.4)	(98.1, **95.2**)	(**99.0**, 82.9)	(**98.3**, **94.1**)
	MPDD	(62.1, 33.0)	(58.7, 29.1)	(76.4, 48.9)	(94.1, 83.2)	(84.0, 61.0)	(15.4, 2.3)	(95.2, 84.2)	(90.2, 69.4)	(**96.5**, **87.0**)	(**95.7**, 84.7)	(**96.5**, **87.9**)
Medical	CVC-ColonDB	(49.5, 15.8)	(49.5, 11.5)	(70.3, 32.5)	(78.4, 64.6)	(77.9, 49.8)	(40.5, 2.6)	(79.1, 69.7)	(83.6, 66.9)	(**81.9**, **71.3**)	(81.5, 63.9)	(**85.0**, **73.3**)
	CVC-ClinicDB	(47.5, 18.9)	(48.5, 12.6)	(51.2, 13.8)	(80.5, 60.7)	(82.1, 55.0)	(34.8, 2.4)	(83.4, **68.8**)	(79.5, 63.1)	(82.9, 67.8)	(**84.6**, 62.3)	(**84.7**, **70.1**)
	Kvasir	(44.6, 17.7)	(45.0, 16.8)	(69.7, 24.5)	(75.0, 36.2)	(72.5, 28.2)	(44.1, 3.5)	(79.1, 38.6)	(76.5, 37.6)	(78.9, 45.6)	(**85.0**, **53.2**)	(**82.1**, **49.9**)
	Endo	(45.2, 15.9)	(46.6, 12.6)	(68.2, 28.3)	(81.9, 54.9)	(80.5, 47.7)	(40.6, 3.9)	(83.1, 59.0)	(78.5, 50.0)	(84.1, 63.6)	(**87.1**, **65.6**)	(**86.8**, **67.6**)
	ISIC	(33.1, 5.8)	(36.0, 7.7)	(83.3, 55.1)	(89.4, 77.2)	(90.6, 79.8)	(51.7, 15.9)	(81.9, 68.9)	(79.8, 57.6)	(**89.7**, 78.4)	(**91.1**, **80.1**)	(**91.1**, **81.6**)
	TN3K	(42.3, 7.3)	(35.6, 5.2)	(70.7, 39.8)	(73.6, 37.8)	(80.9, 50.5)	(34.0, 9.5)	(72.4, 41.0)	(73.9, 44.3)	(**81.5**, **50.4**)	(79.8, 48.5)	(**84.7**, **54.6**)

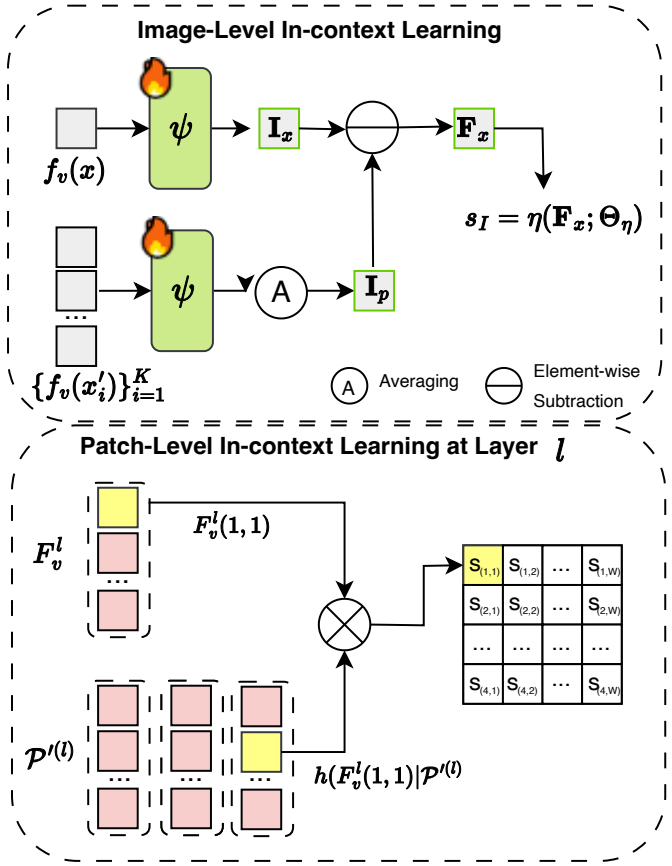
Table 2. Pixel-level ZSAD results (AUROC, PRO) on 14 AD datasets. The best and second-best results are respectively highlighted in **red** and **blue**. Note that medical datasets in Table 1 do not have pixel-level ground truth. Thus, different medical datasets are used here. Detailed breakdown results for MVTecAD, VisA, DAGM, DTD-Synthetic, BTAD, and MPDD can be found in Appendix D.1.

Few-shot Residual Learning & Regression

Our approach: InCTRL

WinCLIP's text prompt – semantic-guided anomaly score





capture the higher-level discrepancies between the query image and the normal image prompts:

$$\mathbf{I}_p = \frac{1}{K}\sum \psi(f_v(x'_k);\Theta_\psi)$$

$$\mathbf{F}_x = \mathbf{I}_x \ominus \mathbf{I}_p$$

$$\mathcal{L}_{IRL} = \frac{1}{N}\sum_{x\in X_{train}} \mathcal{L}_b(\eta(\mathbf{F}_x;\Theta_\eta), y_x)$$

capture fine-grained in-context residuals:

$$\mathbf{M}_x^l(i,j) = 1 - \langle T_x^l(i,j), h(T_x^l(i,j)|\mathcal{P}')\rangle$$

$$\mathbf{M}_x = \frac{1}{n}\sum_{l=1}^{n}\mathbf{M}_x^l$$

Motivation of InCTRL: Instead of modeling dataset-specific information, it's better to model some generalist information that can be transferrable well across diverse data domains. We believe the residual (discrepancy) information between the query sample and given few-shot normal sample prompts is a good choice.

The holistic in-context residual map: $\mathbf{M}_x^+ = \mathbf{M}_x \oplus s_i(x) \oplus s_a(x)$

Zhu, Jiawen, and Guansong Pang. "Toward generalist anomaly detection via in-context residual learning with few-shot sample prompts." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024

Our approach: InCTRL

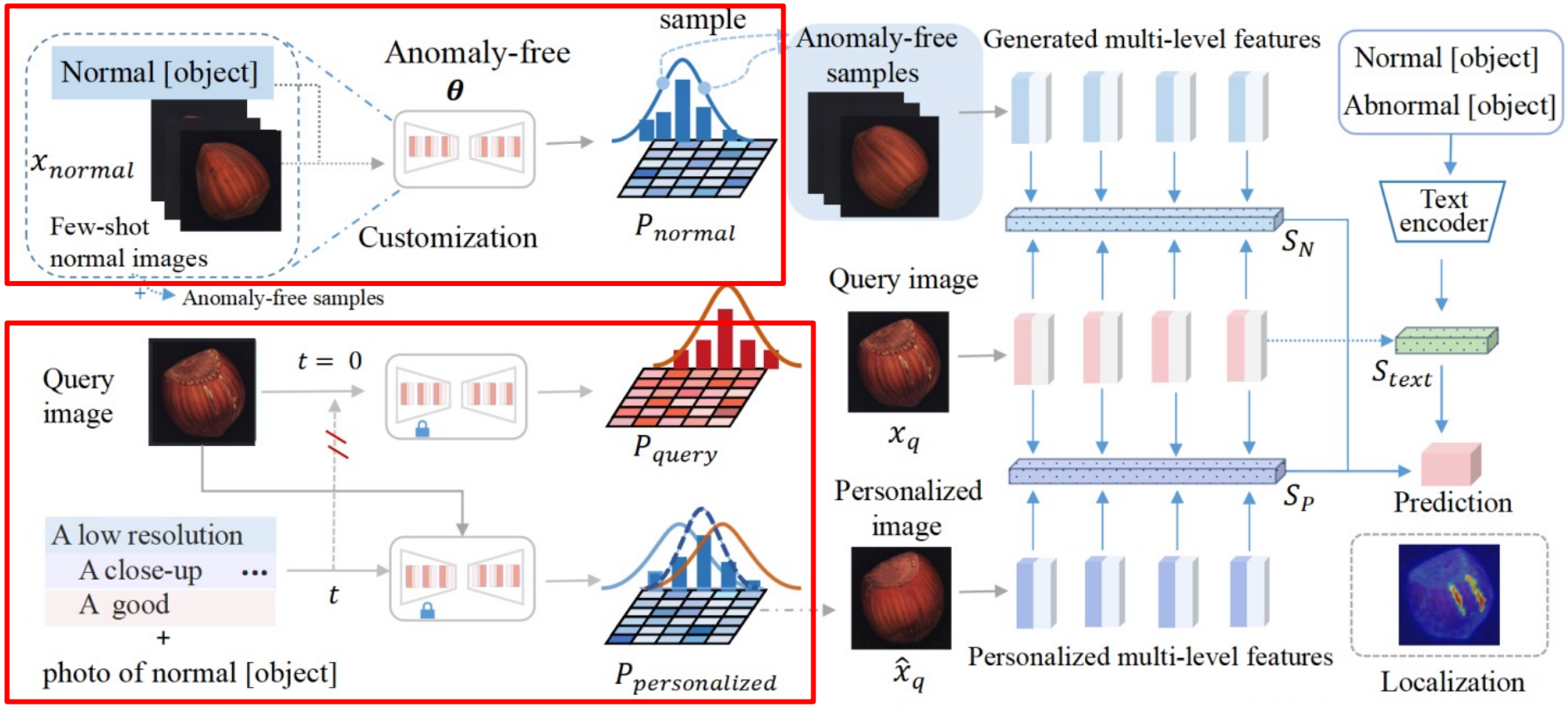
Setup	Methods	Industrial Defects					Medical Anomalies		Semantic Anomalies			
									One-vs-all		Multi-class	
		ELPV	SDD	AITEX	VisA	MVTec AD	BrainMRI	HeadCT	MNIST	CIFAR-10	MNIST	CIFAR-10
Baseline (0-shot)		0.733±0.000	0.946±0.000	0.733±0.000	0.781±0.000	0.912±0.000	0.926±0.000	0.900±0.000	0.678±0.000	0.924±0.000	0.620±0.000	0.900±0.000
2-shot	SPADE	0.517±0.012	0.729±0.041	0.727±0.004	0.795±0.045	0.817±0.054	0.754±0.048	0.645±0.034	0.779±0.024	0.823±0.014	0.595±0.060	0.655±0.042
	PaDiM	0.594±0.083	0.721±0.015	**0.784±0.028**	0.680±0.042	0.785±0.025	0.657±0.122	0.595±0.036	-	-	-	-
	Patchcore	0.716±0.031	0.902±0.006	0.739±0.017	0.817±0.028	0.858±0.034	0.706±0.009	0.736±0.096	0.756±0.004	0.602±0.009	0.603±0.009	0.703±0.008
	RegAD	0.571±0.016	0.499±0.008	0.564±0.072	0.557±0.053	0.640±0.047	0.449±0.129	0.602±0.018	0.525±0.030	0.534±0.005	0.608±0.026	0.695±0.002
	CoOp	0.762±0.011	0.897±0.006	0.687±0.062	0.806±0.023	0.858±0.016	0.725±0.020	0.811±0.003	0.557±0.006	0.527±0.011	0.612±0.007	0.393±0.009
	WinCLIP	**0.726±0.020**	**0.942±0.006**	0.726±0.055	**0.842±0.024**	**0.931±0.019**	**0.934±0.012**	**0.915±0.015**	**0.810±0.008**	**0.925±0.001**	**0.632±0.000**	**0.914±0.005**
	Ours (InCTRL)	**0.839±0.003**	**0.972±0.011**	**0.761±0.029**	**0.858±0.022**	**0.940±0.015**	**0.973±0.027**	**0.929±0.025**	**0.892±0.009**	**0.935±0.002**	**0.635±0.010**	**0.924±0.005**
4-shot	SPADE	0.537±0.013	0.731±0.020	0.718±0.011	0.811±0.040	0.828±0.044	0.759±0.070	0.624±0.012	0.810±0.009	0.836±0.006	0.588±0.041	0.631±0.063
	PaDiM	0.612±0.080	0.742±0.014	**0.787±0.038**	0.735±0.031	0.805±0.018	0.792±0.048	0.622±0.013	-	-	-	-
	Patchcore	0.756±0.073	0.923±0.008	0.733±0.002	0.843±0.025	0.885±0.026	0.794±0.040	0.805±0.006	0.833±0.009	0.639±0.010	0.497±0.044	0.739±0.011
	RegAD	0.596±0.040	0.525±0.027	0.596±0.074	0.574±0.042	0.663±0.032	0.571±0.149	0.522±0.050	0.548±0.053	0.534±0.002	0.596±0.075	0.677±0.161
	CoOp	**0.781±0.002**	0.902±0.006	0.720±0.017	0.818±0.018	0.874±0.017	0.759±0.033	0.860±0.032	0.563±0.004	0.537±0.005	0.618±0.002	0.395±0.008
	WinCLIP	0.754±0.009	**0.943±0.004**	0.764±0.025	**0.858±0.025**	**0.940±0.021**	**0.941±0.002**	**0.912±0.003**	**0.851±0.010**	**0.927±0.001**	**0.632±0.004**	**0.915±0.003**
	Ours (InCTRL)	**0.846±0.011**	**0.975±0.006**	**0.790±0.018**	**0.877±0.019**	**0.945±0.018**	**0.975±0.016**	**0.933±0.013**	**0.902±0.016**	**0.940±0.010**	**0.643±0.007**	**0.928±0.009**
8-shot	SPADE	0.567±0.034	0.741±0.011	0.708±0.006	0.821±0.042	0.840±0.057	0.794±0.039	0.626±0.022	0.829±0.009	0.849±0.006	0.597±0.028	0.656±0.037
	PaDiM	0.724±0.017	0.769±0.037	0.792±0.025	0.768±0.032	0.820±0.016	0.758±0.025	0.661±0.039	-	-	-	-
	Patchcore	**0.837±0.016**	0.925±0.003	0.745±0.002	0.860±0.026	0.922±0.019	0.812±0.016	0.817±0.034	**0.876±0.004**	0.672±0.006	0.526±0.019	0.764±0.004
	RegAD	0.633±0.027	0.594±0.029	0.603±0.062	0.589±0.040	0.674±0.033	0.632±0.079	0.628±0.026	0.547±0.063	0.555±0.008	0.573±0.076	0.587±0.211
	CoOp	0.817±0.012	0.898±0.005	0.769±0.008	0.822±0.021	0.880±0.014	0.755±0.003	0.914±0.027	0.567±0.007	0.542±0.005	0.619±0.004	0.399±0.006
	WinCLIP	0.814±0.010	**0.941±0.001**	**0.796±0.015**	**0.868±0.020**	**0.947±0.025**	**0.944±0.001**	**0.915±0.008**	0.867±0.007	**0.928±0.001**	**0.641±0.004**	**0.916±0.003**
	Ours (InCTRL)	**0.872±0.013**	**0.978±0.006**	**0.806±0.036**	**0.887±0.021**	**0.953±0.013**	**0.983±0.012**	**0.936±0.008**	**0.920±0.003**	**0.945±0.002**	**0.646±0.003**	**0.934±0.008**

Setup	Methods	Industrial Defects					Medical Anomalies		Semantic Anomalies			
									One-vs-all		Multi-class	
		ELPV	SDD	AITEX	VisA	MVTec AD	BrainMRI	HeadCT	MNIST	CIFAR-10	MNIST	CIFAR-10
Baseline (0-shot)		0.855±0.000	0.886±0.000	0.552±0.000	0.812±0.000	0.957±0.000	0.988±0.000	0.970±0.000	0.940±0.000	0.990±0.000	0.606±0.000	0.852±0.000
2-shot	SPADE	0.618±0.007	0.366±0.105	0.470±0.008	0.818±0.031	0.922±0.023	0.952±0.009	0.851±0.022	**0.965±0.004**	0.971±0.003	**0.615±0.068**	0.502±0.035
	PaDiM	0.707±0.058	0.337±0.008	**0.529±0.034**	0.719±0.027	0.890±0.015	0.902±0.046	0.876±0.017	-	-	-	-
	Patchcore	0.840±0.031	0.676±0.003	0.378±0.008	0.841±0.023	0.939±0.012	0.921±0.017	0.913±0.002	0.956±0.001	0.926±0.002	0.482±0.025	0.574±0.015
	RegAD	0.679±0.005	0.173±0.019	0.275±0.035	0.614±0.037	0.837±0.034	0.872±0.065	0.854±0.009	0.913±0.006	0.909±0.003	0.612±0.013	0.672±0.008
	CoOp	0.841±0.020	0.543±0.004	0.443±0.050	0.835±0.019	0.922±0.007	0.923±0.002	0.937±0.014	0.926±0.003	0.911±0.002	0.607±0.009	0.371±0.013
	WinCLIP	**0.849±0.010**	**0.865±0.004**	0.500±0.043	**0.859±0.021**	**0.965±0.007**	**0.989±0.003**	**0.975±0.012**	0.963±0.001	**0.990±0.001**	0.614±0.005	**0.876±0.016**
	Ours (InCTRL)	**0.913±0.008**	**0.917±0.009**	**0.519±0.022**	**0.877±0.016**	**0.969±0.004**	**0.994±0.013**	**0.981±0.013**	**0.975±0.004**	**0.992±0.000**	**0.618±0.012**	**0.899±0.010**
4-shot	SPADE	0.627±0.011	0.385±0.018	0.451±0.031	0.826±0.024	0.924±0.015	0.958±0.017	0.854±0.016	0.966±0.008	0.973±0.002	**0.611±0.053**	0.487±0.047
	PaDiM	0.724±0.067	0.351±0.012	**0.540±0.053**	0.758±0.018	0.909±0.013	0.956±0.011	0.890±0.011	-	-	-	-
	Patchcore	**0.871±0.042**	0.703±0.013	0.377±0.001	0.860±0.016	0.950±0.013	0.945±0.017	0.941±0.009	**0.972±0.002**	0.934±0.003	0.504±0.025	0.606±0.010
	RegAD	0.688±0.018	0.176±0.003	0.294±0.031	0.628±0.034	0.846±0.026	0.900±0.041	0.810±0.028	0.916±0.013	0.908±0.001	0.522±0.085	0.681±0.127
	CoOp	0.867±0.003	0.594±0.014	0.454±0.014	0.842±0.016	0.924±0.008	0.932±0.013	0.957±0.017	0.929±0.002	0.915±0.003	0.611±0.003	0.374±0.012
	WinCLIP	0.864±0.004	**0.868±0.003**	0.513±0.017	**0.875±0.023**	**0.968±0.008**	**0.990±0.001**	**0.974±0.002**	0.971±0.002	**0.990±0.000**	**0.611±0.011**	**0.882±0.009**
	Ours (InCTRL)	**0.916±0.009**	**0.924±0.015**	**0.548±0.016**	**0.902±0.027**	**0.972±0.006**	**0.994±0.013**	**0.984±0.011**	**0.980±0.007**	**0.992±0.004**	**0.620±0.004**	**0.901±0.020**
8-shot	SPADE	0.641±0.018	0.394±0.024	0.427±0.008	0.844±0.031	0.930±0.016	0.962±0.014	0.860±0.019	0.974±0.002	0.976±0.001	0.613±0.035	0.515±0.024
	PaDiM	0.798±0.014	0.384±0.045	0.555±0.031	0.781±0.024	0.927±0.012	0.946±0.007	0.896±0.009	-	-	-	-
	Patchcore	**0.915±0.007**	0.708±0.009	0.389±0.003	0.873±0.022	0.962±0.013	0.957±0.007	0.931±0.006	**0.979±0.001**	0.942±0.002	0.530±0.037	0.635±0.019
	RegAD	0.696±0.015	0.246±0.031	0.314±0.036	0.643±0.032	0.855±0.021	0.908±0.013	0.881±0.014	0.919±0.018	0.911±0.001	0.566±0.048	0.558±0.159
	CoOp	0.905±0.008	0.578±0.001	0.514±0.003	0.848±0.020	0.933±0.007	0.927±0.007	0.965±0.018	0.937±0.004	0.920±0.003	0.610±0.001	0.376±0.003
	WinCLIP	0.897±0.007	**0.865±0.001**	**0.562±0.024**	**0.880±0.021**	**0.973±0.009**	**0.991±0.000**	**0.975±0.003**	0.974±0.001	**0.990±0.000**	**0.616±0.006**	**0.887±0.006**
	Ours (InCTRL)	**0.926±0.006**	**0.925±0.011**	**0.561±0.034**	**0.904±0.025**	**0.977±0.006**	**0.996±0.003**	**0.985±0.005**	**0.989±0.001**	**0.994±0.001**	**0.622±0.008**	**0.912±0.005**

Zhu, Jiawen, and Guansong Pang. "Toward generalist anomaly detection via in-context residual learning with few-shot sample prompts." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024

One-to-Normal

① uses the few-shot normal samples to fine-tune the anomaly-free customized model using Dreambooth.



② perform one-to-normal personalization of the query image, obtaining the personalized image

One-to-normal personalization **Triplet contrastive anomaly inference**

Li Y, Zhang S, Li K, Lao Q. One-to-normal: Anomaly personalization for few-shot anomaly detection. Advances in Neural Information Processing Systems. 2024

Limitation of the few-shot learning

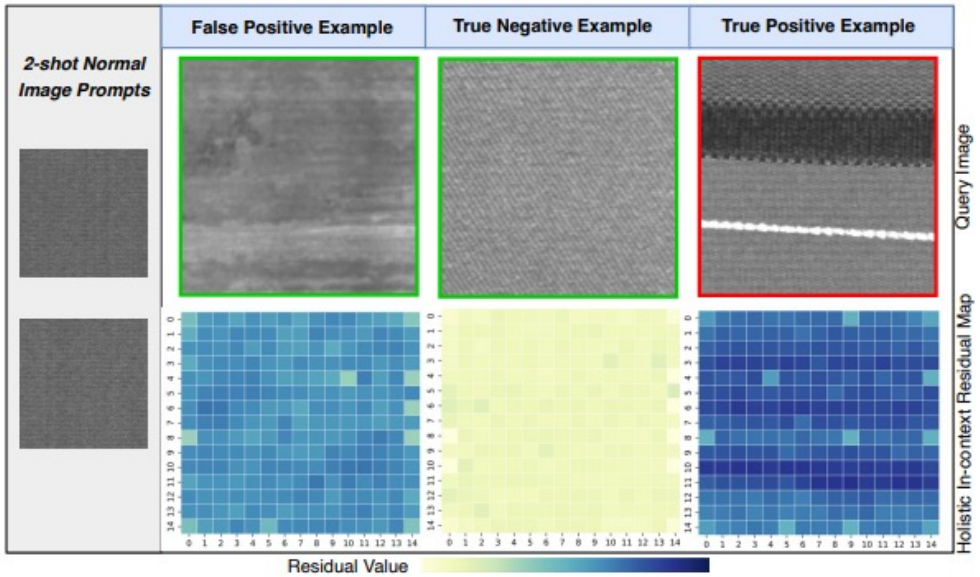


Figure 3. Visualization of query images x_t and their holistic in-context residual maps $\mathbf{M}_{x_t}^{+}$. Green and Red frames indicate normal and abnormal images respectively. Deeper colors in the residual maps represent larger residual values.

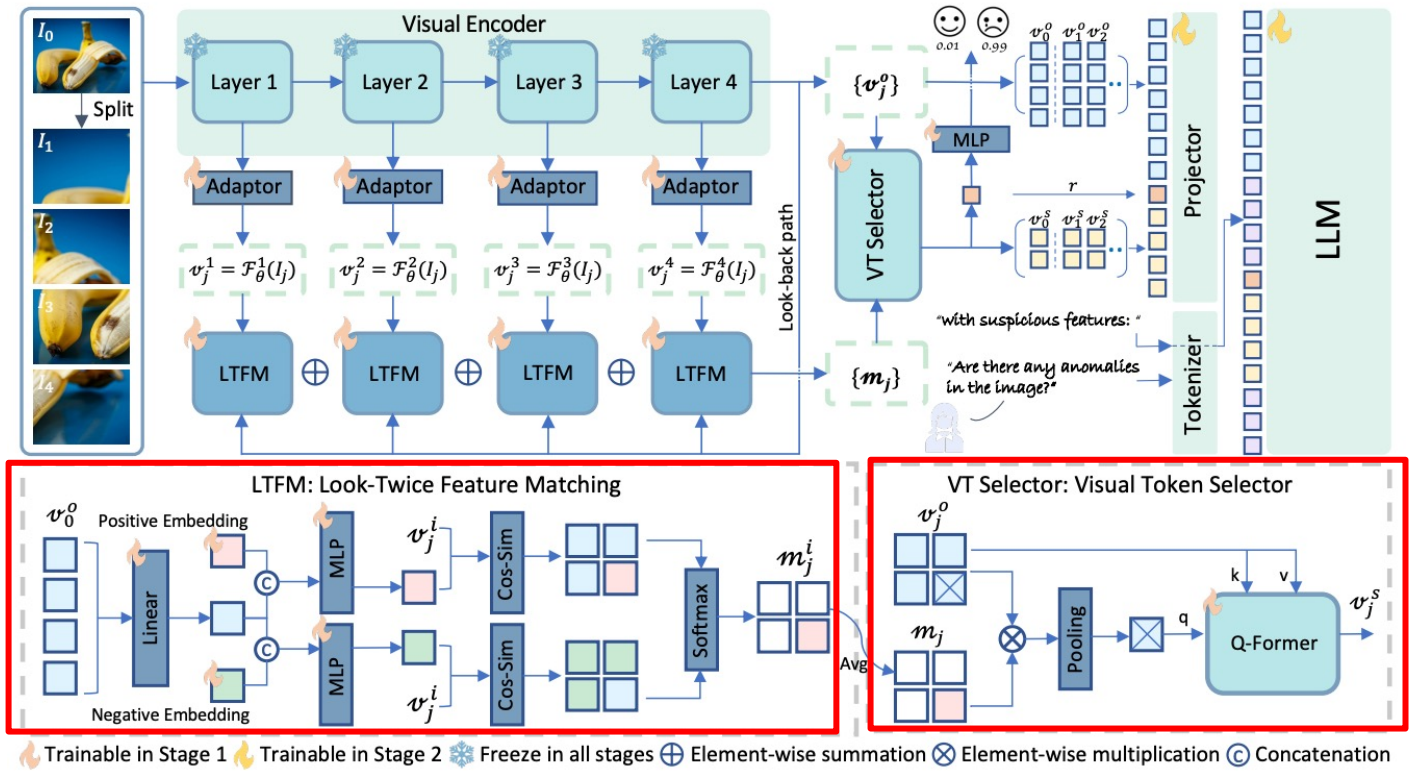
- These model may not capture the full diversity of normal patterns based on few-shot sample prompts, leading to false positives. As shown in the visualization from InCTRL:

1. in the first case, a query image with different texture from the normal samples is mistakenly marked as abnormal, because the residual is large.

2. However, when evaluated on the normal example with visually similar textual, the model correctly identifies it as normal.

Reasoning & Multimodal Integration

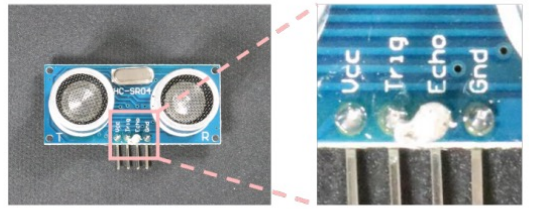
Anomaly-OV



LTFM: recognizes and localizes suspicious visual tokens by comparing visual features with their visual-conditional text features.

VT Selector: refines and aggregates the suspicious tokens identified by LTFM. It multiplies each visual token with its significance score from LTFM, then applies spatial pooling, and uses a Q-Former to select the most critical and informative tokens for the LLM to do later reasoning.

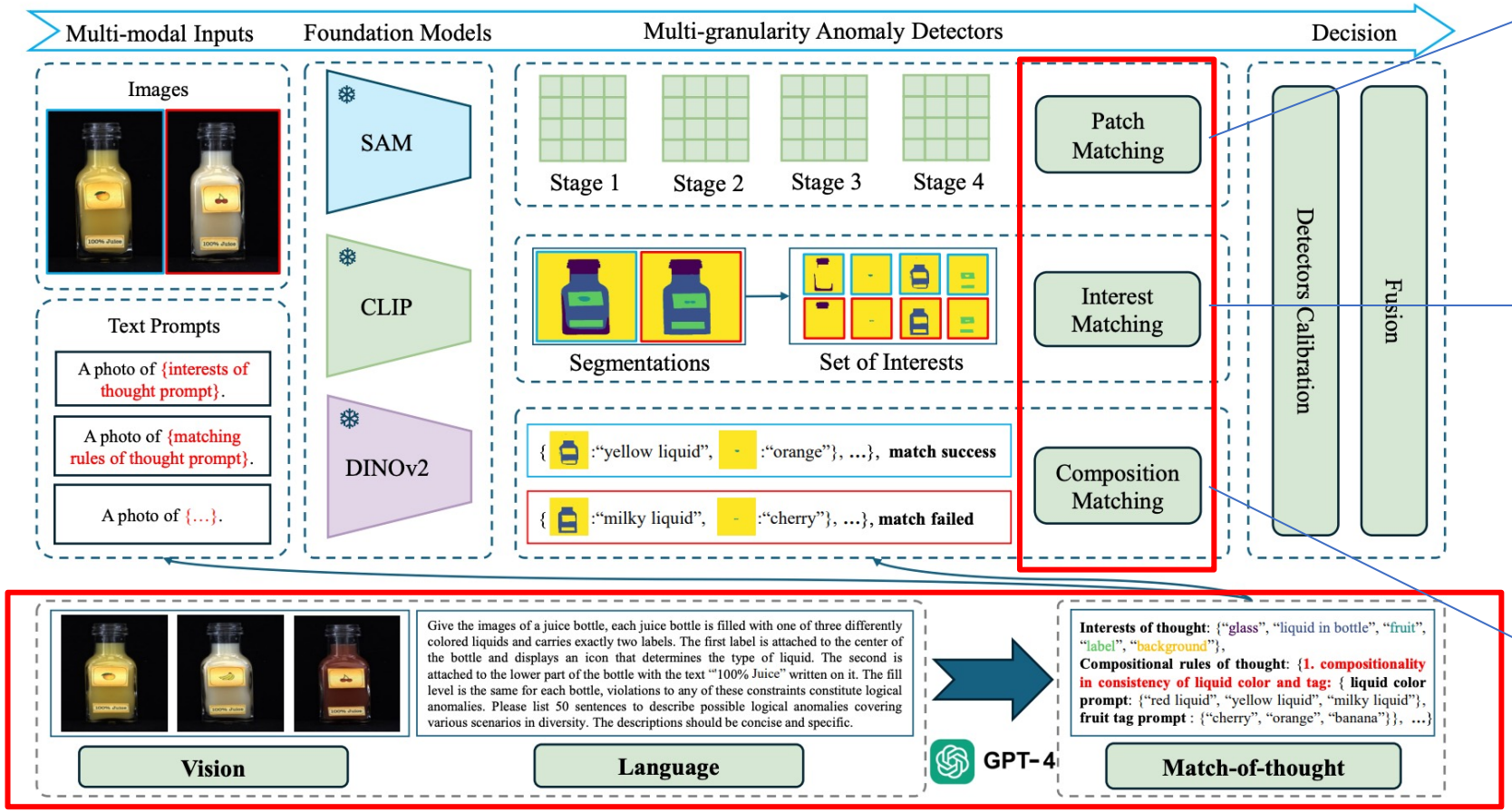
Motivation of Anomaly-OV: It aims to introduce an auxiliary specialist or expert model designed to guide the generalist in selecting and utilizing critical visual tokens

PCB Example: Excessive Solder	
	
User	Are there any anomalies visible in the image?
GPT-4o	The image shows an HC-SR04 ultrasonic sensor. There don't appear to be any visible anomalies with the sensor itself. The components and pins seem intact and properly aligned.
LLaVA-OV	The image shows an ultrasonic sensor module, specifically the HC-SR04 model. There are no visible anomalies in the image; it appears to be a standard representation of this type of sensor module.
Ours	Yes, there is a noticeable anomaly in the image: a short circuit on the PCB, indicated by a white, crusty residue near the 'Echo' pin.

Xu J, Lo SY, Safaei B, Patel VM, Dwivedi I. Towards zero-shot anomaly detection and reasoning with multimodal large language models. InProceedings of the Computer Vision and Pattern Recognition Conference 2025

LogSAD

It performs **multi-granularity matching** between the query sample and anomaly-free memory bank



Low-level structural anomaly detection
based on Patch feature from CLIP and DINOv2

Mid-level semantic consistency
based on averaged patch features extracted from SAM and CLIP guided by MoT to segment out 'interested' patch such as liquid, label, fruit, cap.

High-level logical reasoning
checks whether visual–textual relationships comply with predefined logic rules.

It uses GPT-4 to automatically generate text prompts that describe objects and their compositional rules.

Zhang J, Wang G, Jin Y, Huang D. Towards Training-free Anomaly Detection with Vision and Language Foundation Models. InProceedings of the Computer Vision and Pattern Recognition Conference 2025

Pros and Cons of Different FM-driven AD Paradigms

Categories	Pros	Cons
Simple Training-free & Projection Layer	- No training cost; quick adaptation to new domains. - Fully zero-shot setup.	- Severe domain gap limits generalization. - Cannot adapt to domain-specific features or anomaly patterns.
Text Prompt Tuning	- Improves alignment between text and image spaces in downstream tasks	- Only improves alignment between text and image spaces. - Hand-crafted prompts lack flexibility. - Learnable prompts may misalign with real semantics. - Hard to encode complex or professional anomaly-related knowledge.
Few-shot Residual Learning & Regression	- Leverages limited normal data for adaptation. - Better captures fine-grained deviations via residual learning.	- Few-shot limitation: difficult to model normal patterns reliably. - Sensitive to intra-class variance and dataset quality.
Reasoning & Multimodal Integration	- Enables high-level semantic understanding via multimodal reasoning. - Expands scope beyond image/pixel-level AD.	- Scalability: integrating multiple FMs is costly and complex. - Annotation cost: reasoning tasks need detailed labels. - Trustworthiness: LLM evaluation is not reliable, but human evaluation is expensive.

Thanks for Listening!